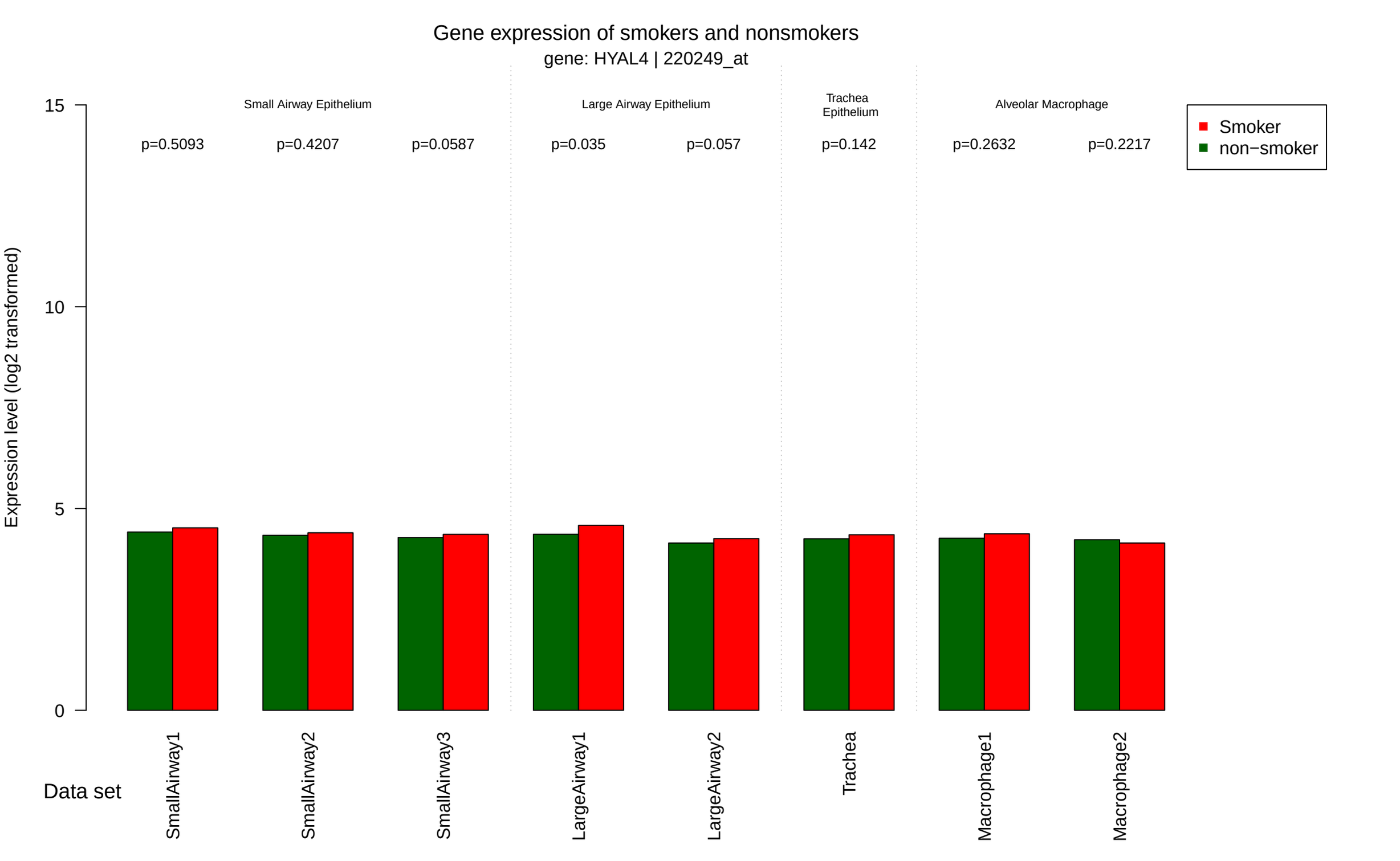
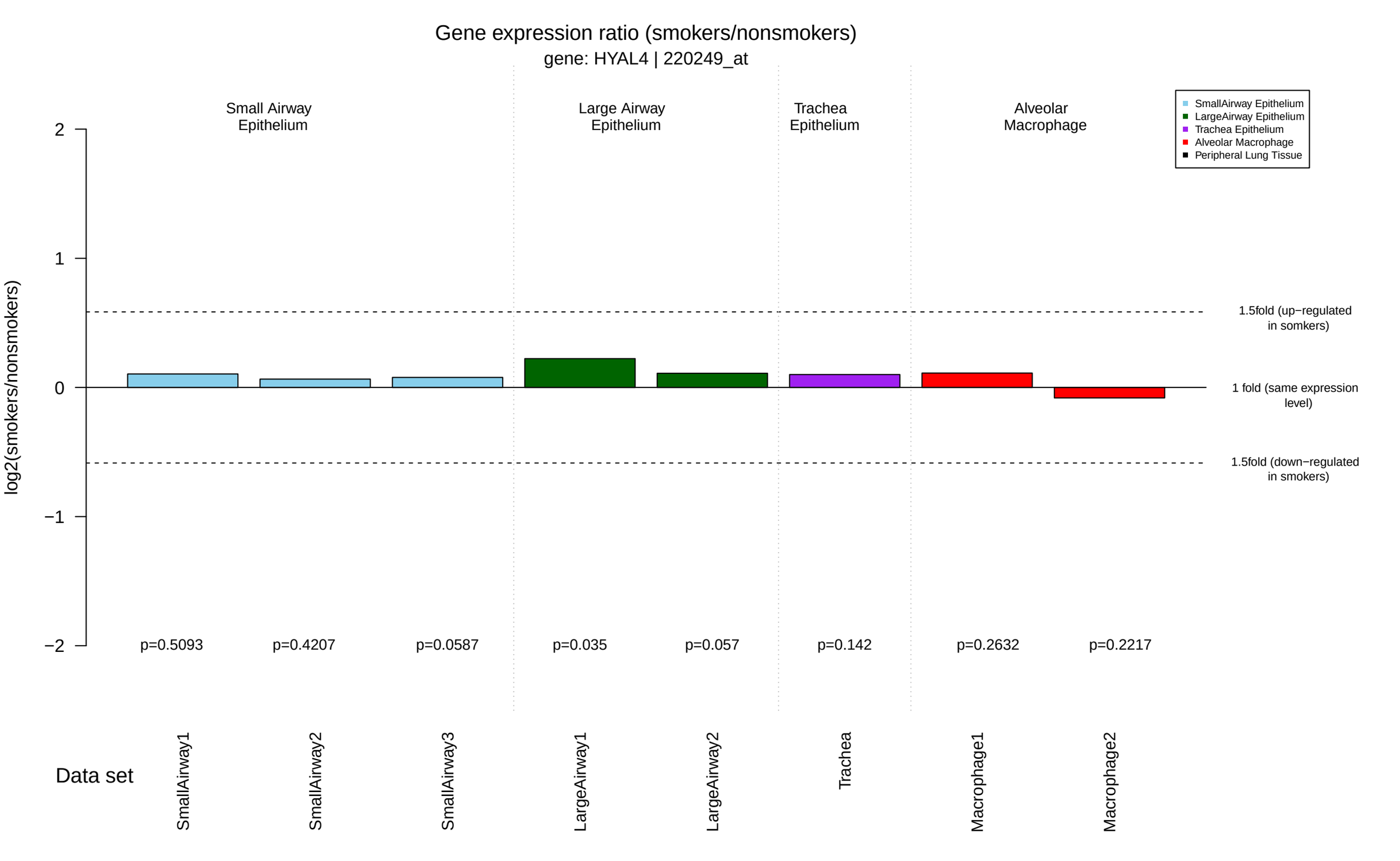
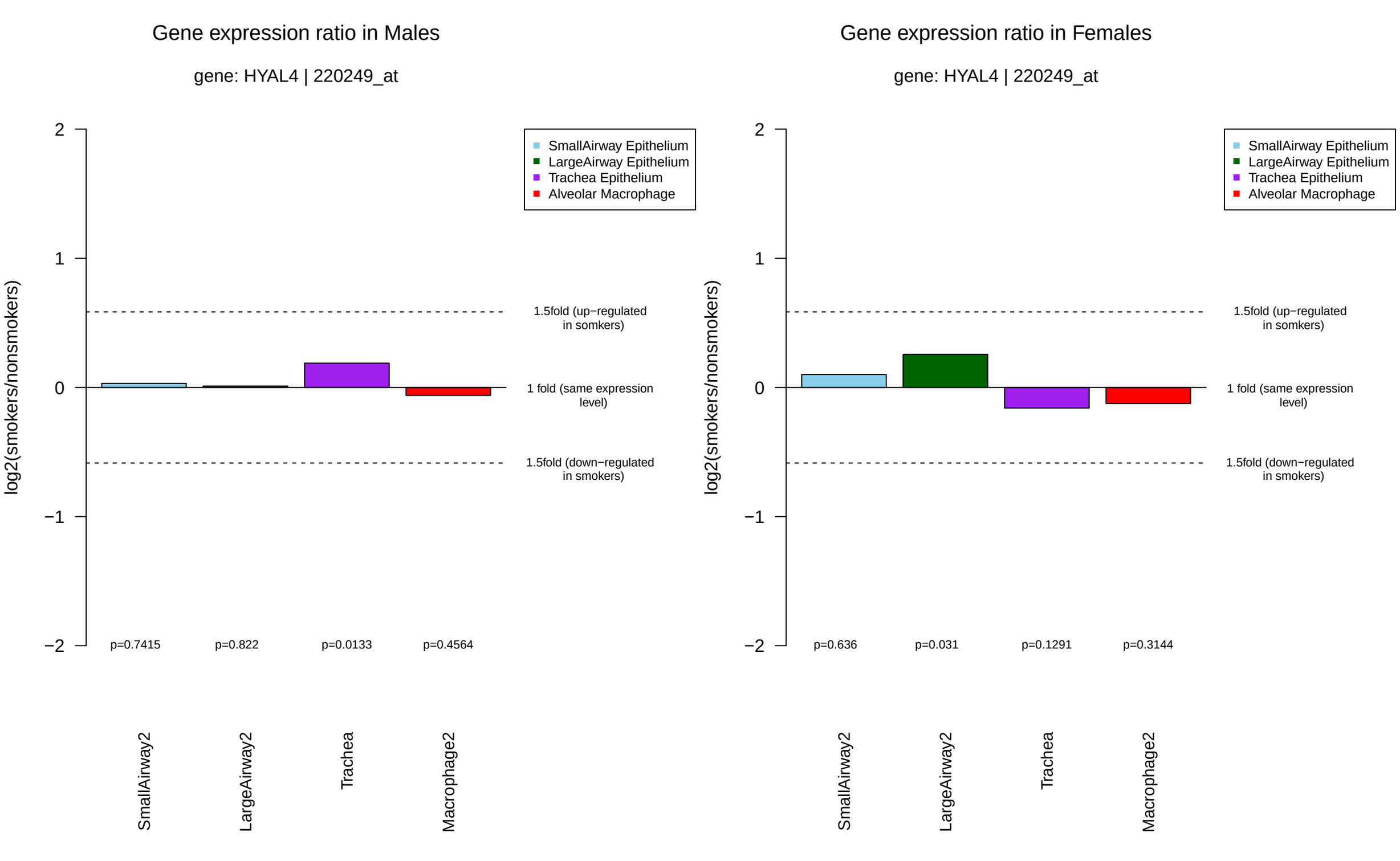
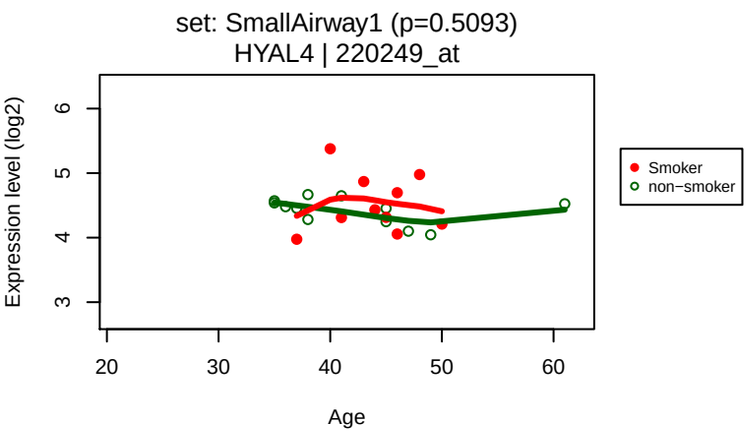
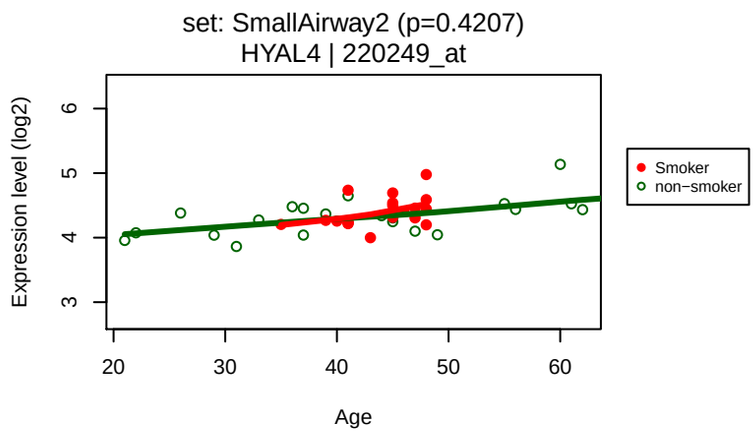
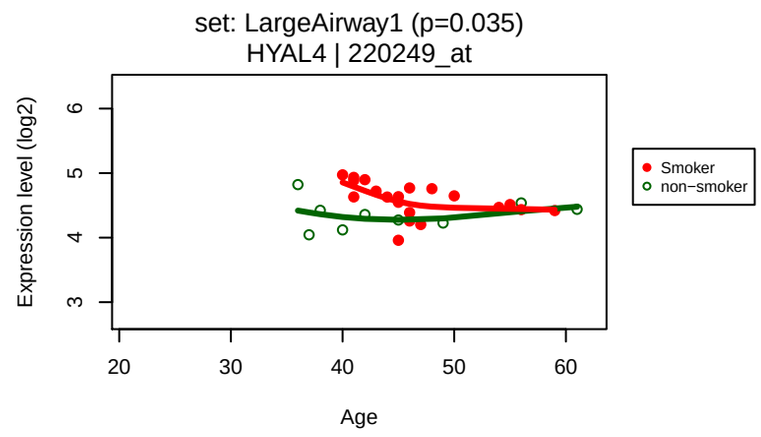

Gene expression of smokers and nonsmokers
gene: HYAL4 | 220249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5093
p=0.4207
p=0.0587
p=0.035
p=0.057
p=0.142
p=0.2632
p=0.2217
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

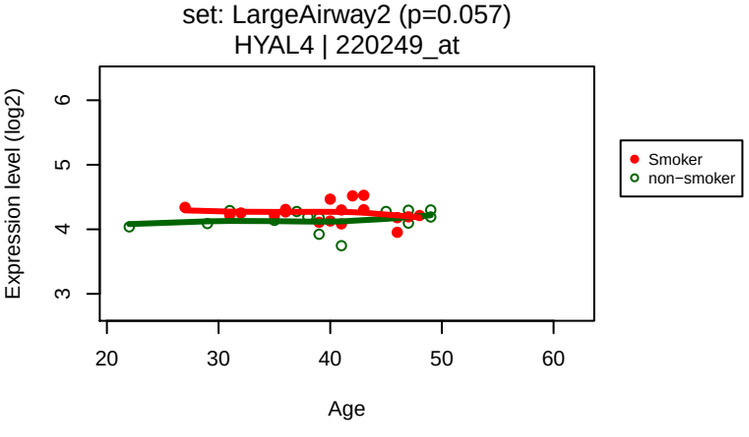

Gene expression ratio (smokers/nonsmokers)
gene: HYAL4 | 220249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5093
p=0.4207
p=0.0587
p=0.035
p=0.057
p=0.142
p=0.2632
p=0.2217
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

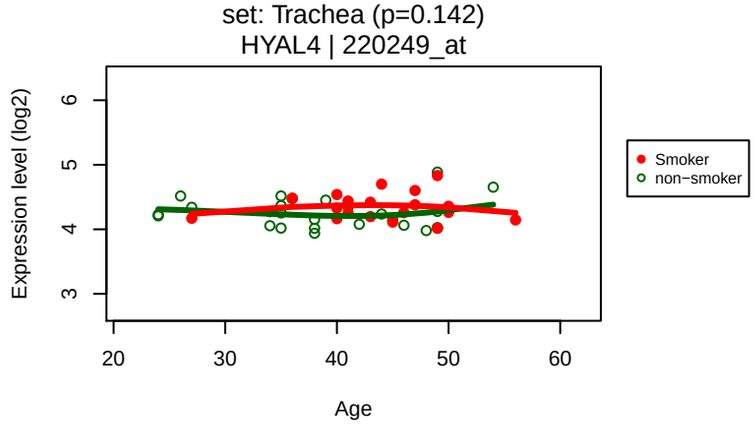

Gene expression ratio in Males
gene: HYAL4 | 220249_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7415
p=0.822
p=0.0133
p=0.4564
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HYAL4 | 220249_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.636
p=0.031
p=0.1291
p=0.3144
SmallAirway2
LargeAirway2
Trachea
Macrophage2

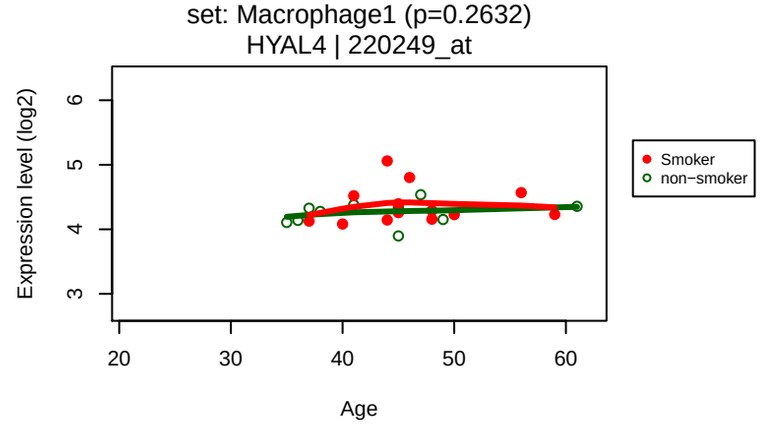

set: SmallAirway1 (p=0.5093)
HYAL4 | 220249_at
Expression level (log2)
Age
Smoker
non−smoker

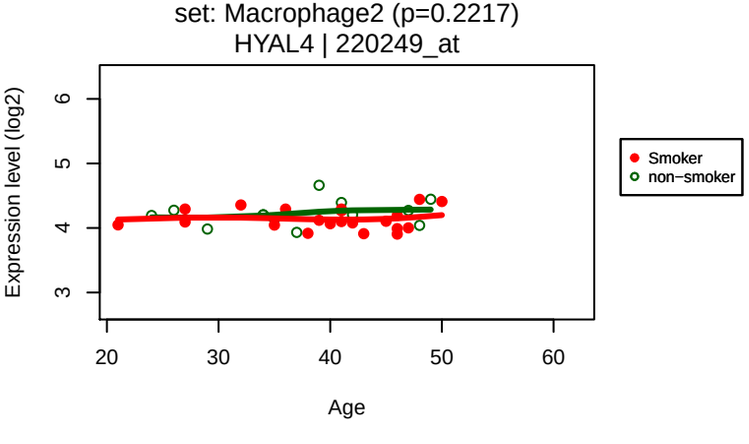

set: SmallAirway2 (p=0.4207)
HYAL4 | 220249_at
Expression level (log2)
Age
Smoker
non−smoker

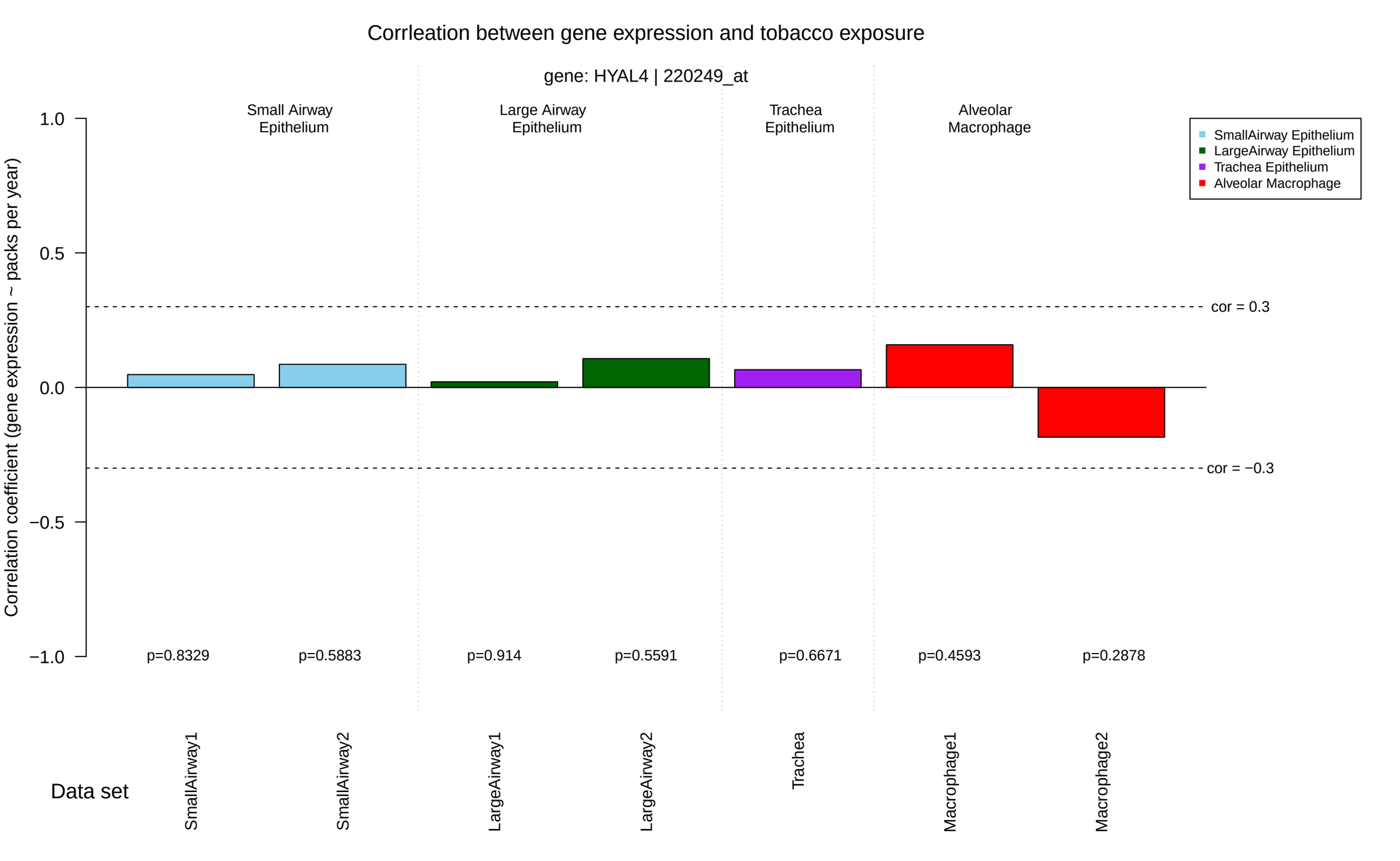

set: LargeAirway1 (p=0.035)
HYAL4 | 220249_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.057)
HYAL4 | 220249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.142)
HYAL4 | 220249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2632)
HYAL4 | 220249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2217)
HYAL4 | 220249_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HYAL4 | 220249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8329
p=0.5883
p=0.914
p=0.5591
p=0.6671
p=0.4593
p=0.2878
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2